PROJECT 3: Active Directory + Security Hardening

Description

In this project, I will be working with Windows Server to configure Active Directory (AD) — the system used to manage users and computers in virtually every enterprise — and then apply security hardening techniques using Group Policy. This project directly mirrors what IT and security teams do in real corporate environments every single day.

Why This Project Matters

Active Directory is arguably the most widely used identity and access management system in the world. This project demonstrates knowledge of authentication, authorization, least privilege, and Group Policy — concepts central to Active Directory, networking, and cybersecurity

📚 Core Concepts: Identity, Access, and Security Concepts Used in This Project
• Active Directory (AD)
• Domain Controller (DC)
• Organizational Unit (OU)
• Group Policy Object (GPO)
• Least Privilege
• Authentication vs. Authorization
• Account Lockout Policy
• Password Policy
• Security Auditing / Event Logs

Tools Used

Tool / Software	Purpose
Windows Server 2022 (Eval)	Server OS to run Active Directory
Windows 10/11 VM	Client workstation to join the domain
VirtualBox	From Project 1 — already installed
Microsoft RSAT Tools	Remote management of AD from a client

Project Layout:

1. In VirtualBox, created a new VM for the Domain Controller and named it DC-01.
2. For the Active Directory Domain Controller, Windows 2022 64-bit was used. Network adapter set to Internal Network LAN 1, promiscuous mode set to Allow All.
3. Installed using the Windows Server 2022 Standard Evaluation (Desktop Experience), password was set.
4. Manually assigned a static IP address to the device using PowerShell:
    a. First we had to get the interface information for the ethernet adapter:

Get-NetIPConfiguration

    b. After obtaining the interface information, I used the following command to assign the IP, subnet mask, and gateway for the device:

New-NetIPAddress -InterfaceIndex 12 -IPAddress 192.168.10.5 -PrefixLength 24 -DefaultGateway 192.168.10.1

    c. I also assigned the DNS using the following command:

Set-DnsClientServerAddress -InterfaceIndex 12 -ServerAddresses 127.0.0.1

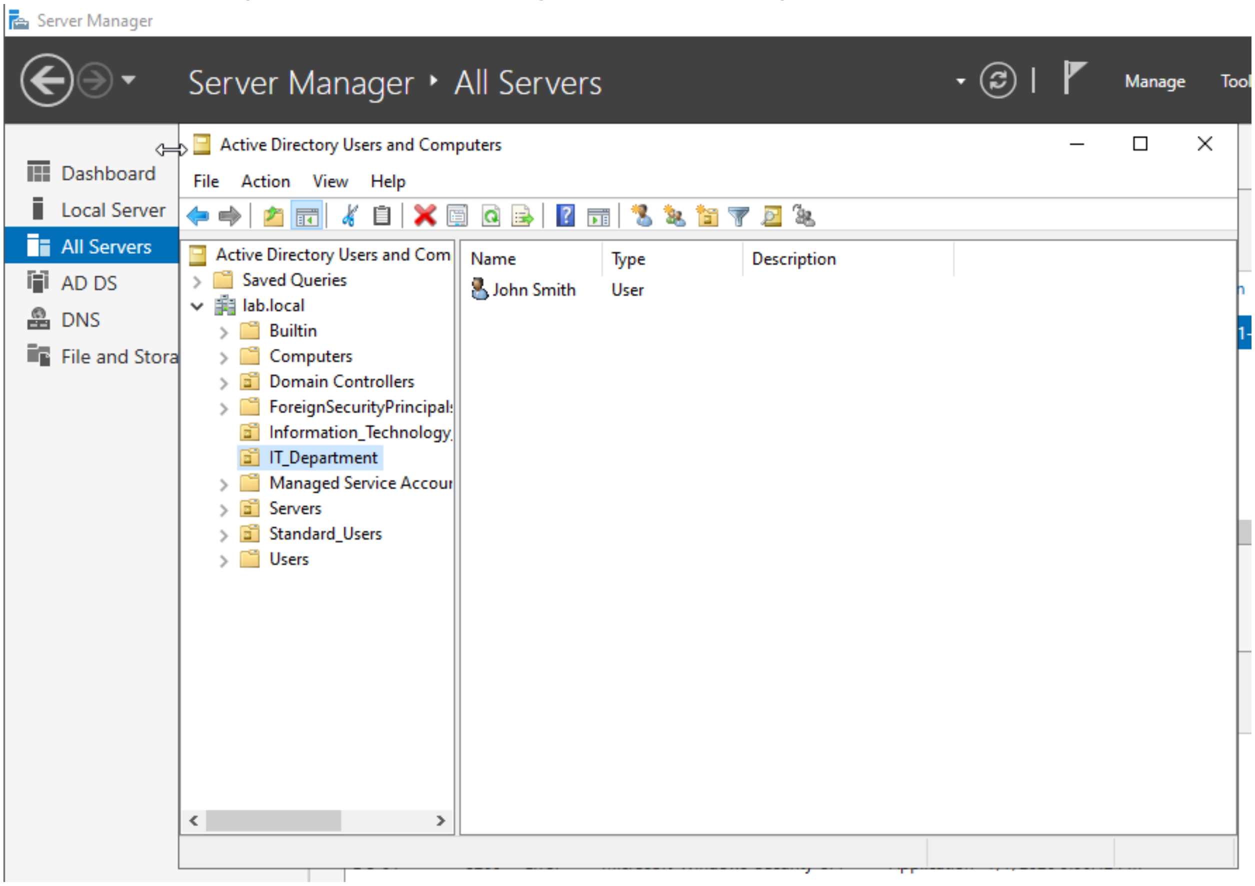
DC-01 Domain Controller Network Configuration:

```
Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
   Physical Address. . . . . . . . . : 08-00-27-B9-93-F0
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::e8f6:ba30:19fd:face%2(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.10.5(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1
   DHCPv6 IAID . . . . . . . . . . . : 101187623
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-DD-CA-16-08-00-27-B9-93-F0
   DNS Servers . . . . . . . . . . . : ::1
                                       127.0.0.1
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

5. Added Roles and Features to the server, selected features in the Active Directory Domain Services.
6. Promoted the server to domain controller and named the root domain as lab.local
7. Forest and Domain functional level set to Windows Server 2016

8. Created Organizational Units using the Server Manager Tools on the lab.local domain:



The following OUs were created: IT_Department, Standard_Users, and Servers

9. Users were created for each OU, passwords assigned.
10. Created a Windows 11 client VM to simulate one of the users we created. Began by installing Windows 11 64 bit, set the Adapter to Internal Network LAN 1, same as the DC-01 VM.
11. After installing Windows 11, went ahead and assigned static IP address, subnet mask, gateway, and DNS on the Client VM:

New-NetIPAddress -InterfaceIndex 12 -IPAddress 192.168.10.20 -PrefixLength 24 -DefaultGateway 192.168.10.1

Set-DnsClientServerAddress -InterfaceIndex 12 -ServerAddresses 192.168.10.5

DNS was set to the IP of the Domain Controller

Client-VM Network Configuration:

```
Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
   Physical Address. . . . . . . . . : 08-00-27-82-A4-1C
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::9f09:9aaa:c956:1b97%12(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.10.20(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.10.1
   DHCPv6 IAID . . . . . . . . . . . : 101187623
   DHCPv6 Client DUID. . . . . . . . : 00-01-01-00-31-DD-95-F3-08-00-27-82-A4-1C
   DNS Servers . . . . . . . . . . . : 192.168.10.5
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

12. To verify connectivity with the domain controller, ping test was done to dc-01.lab.local:

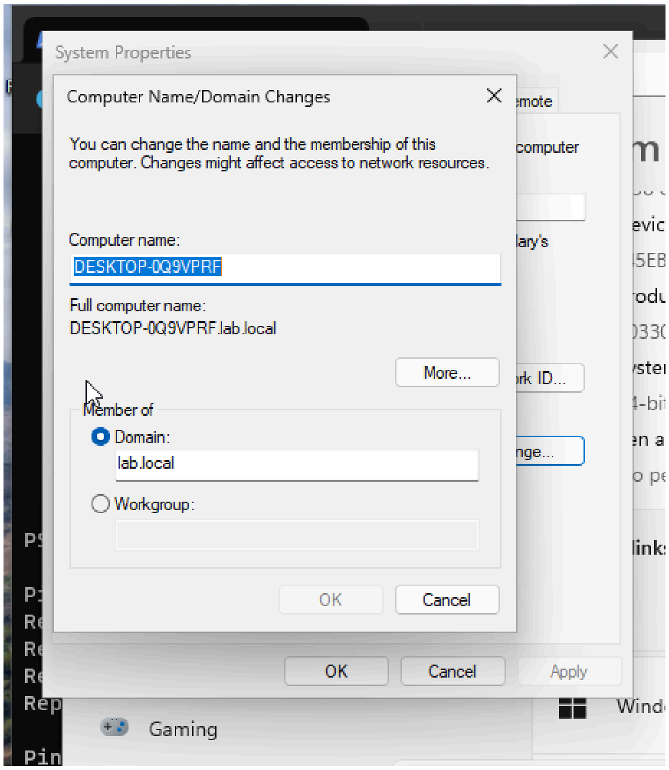
```
PS C:\Users\jsmith> ping dc-01.lab.local

Pinging dc-01.lab.local [192.168.10.5] with 32 bytes of data:
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time=1ms TTL=128
Reply from 192.168.10.5: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping successful

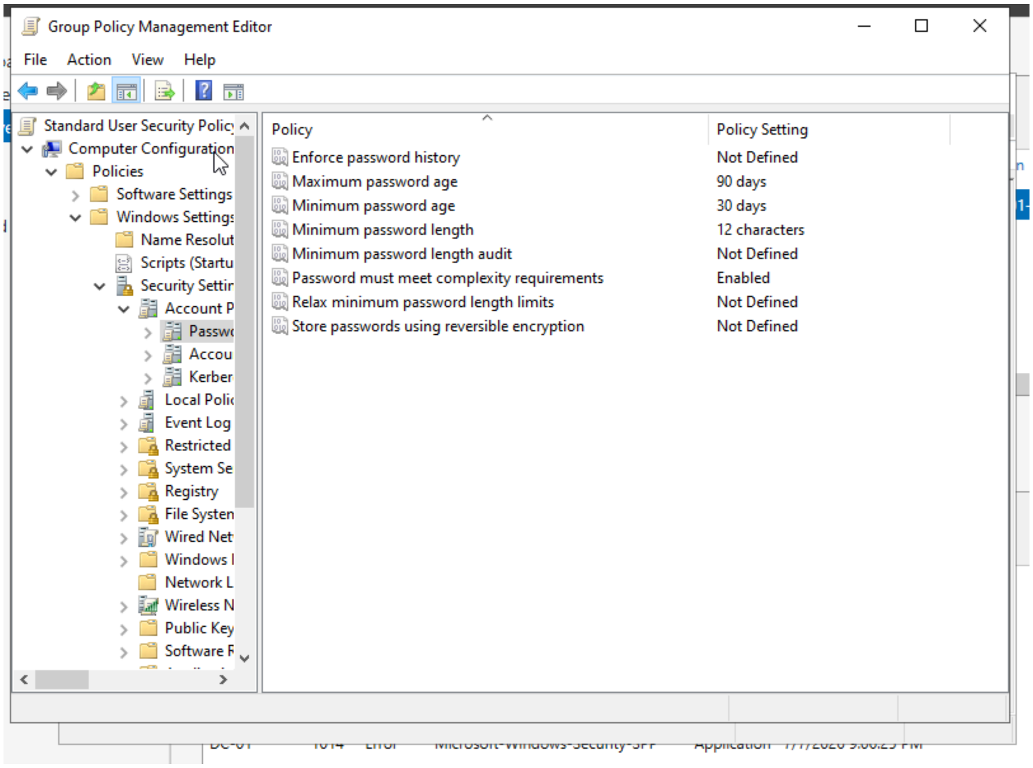
13. On Client VM, navigated to System Settings to add it to the lab.local domain



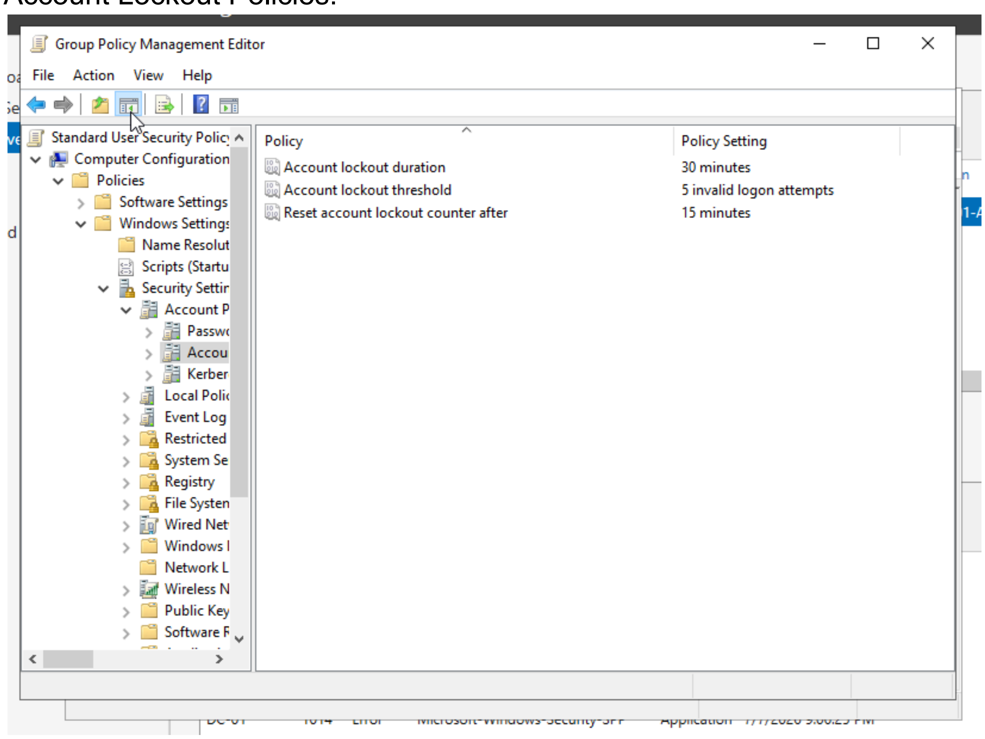
14. Restarted the Client-VM. Upon restart selected Other User and used the credentials for the user set in the IT_Department. Login successful, successfully authenticated with the Domain Controller

15. Group Policies were set for each OU. Standard_Users were set with the following policies: \

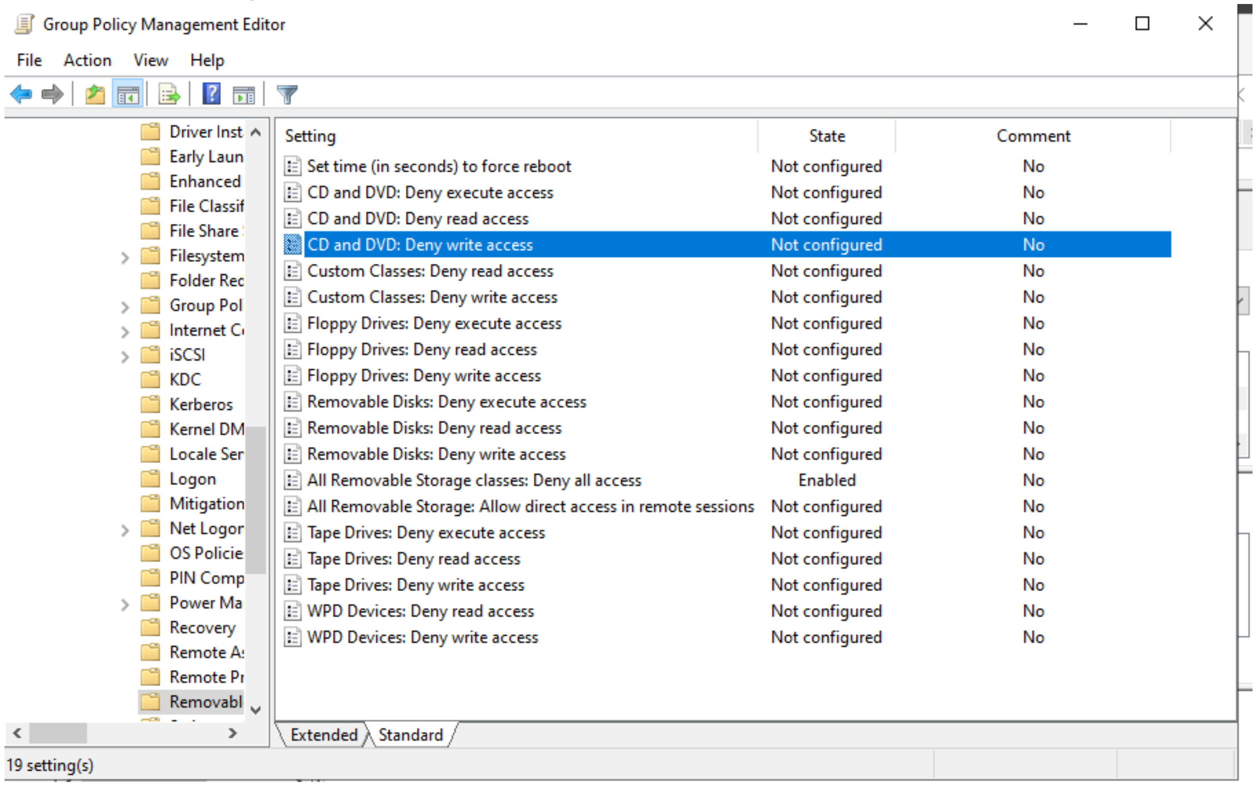
Password Policies:



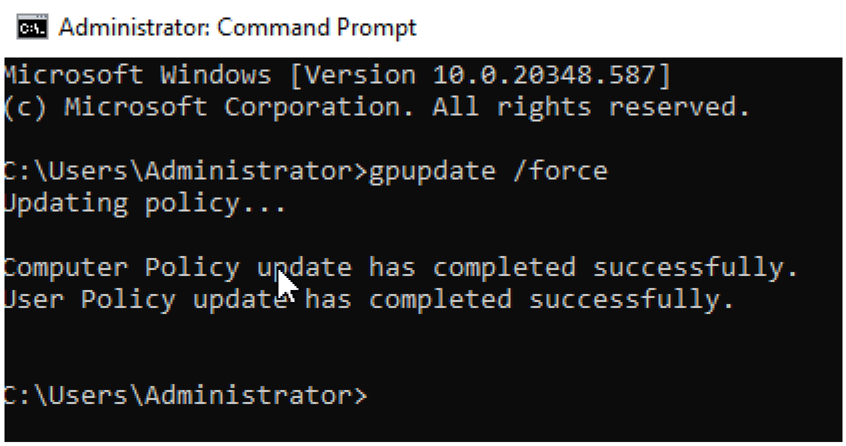
Account Lockout Policies:



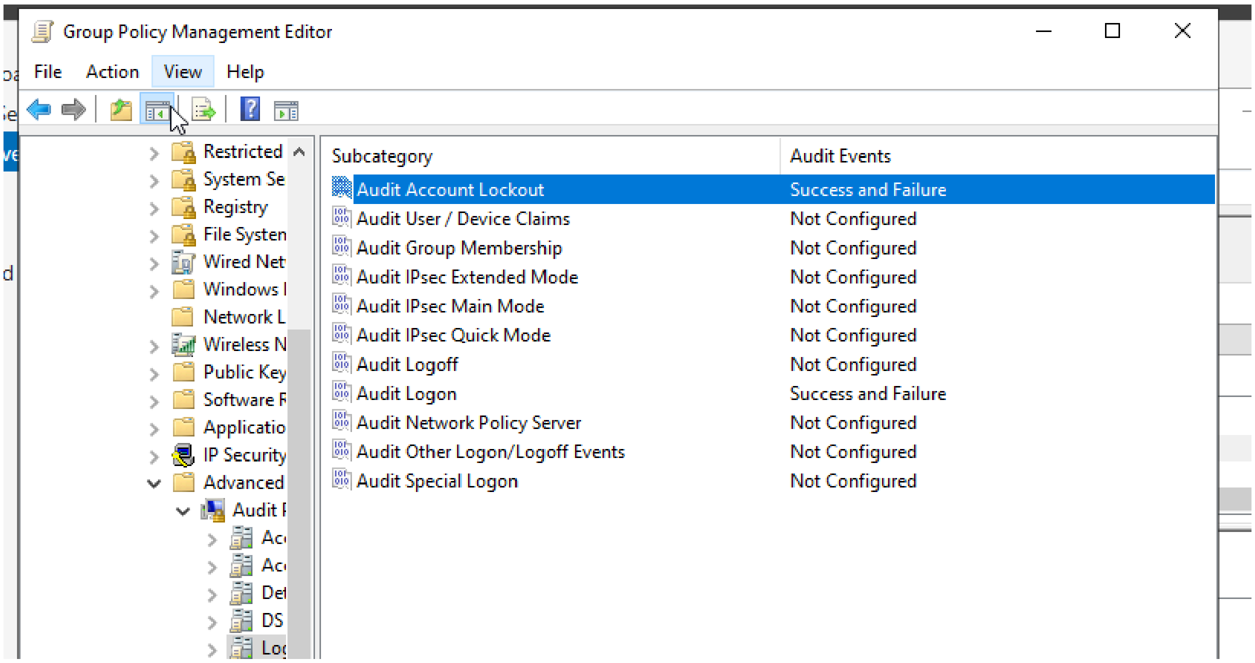
Disable USB Storage:



gpupdate /force was used to enforce latest policy changes

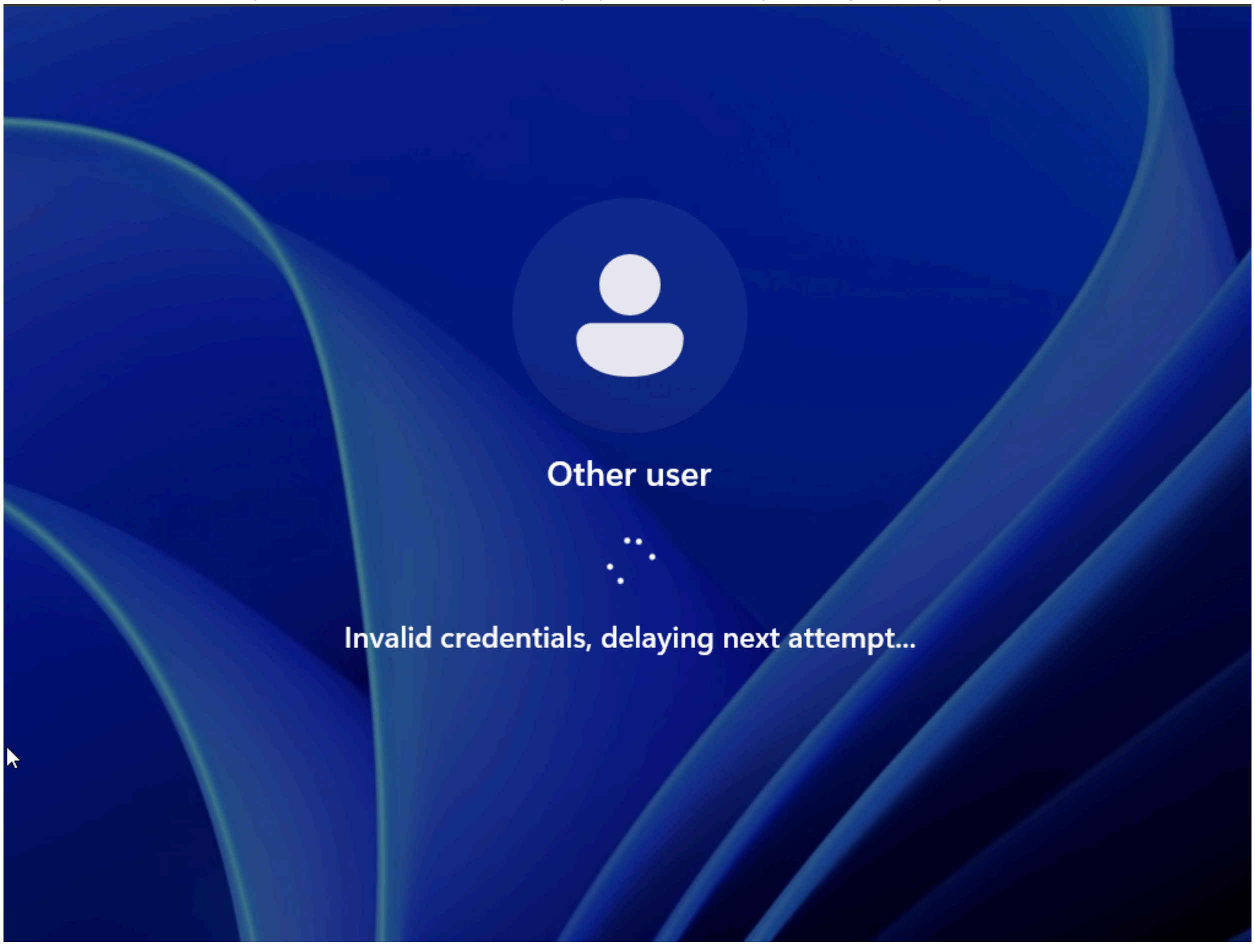


16. Enabled security auditing by going into the Security Policy GPO for Standard Users and Enabling the Audit Logon and Audit Account Lockout option for both successful and failed logon attempts:

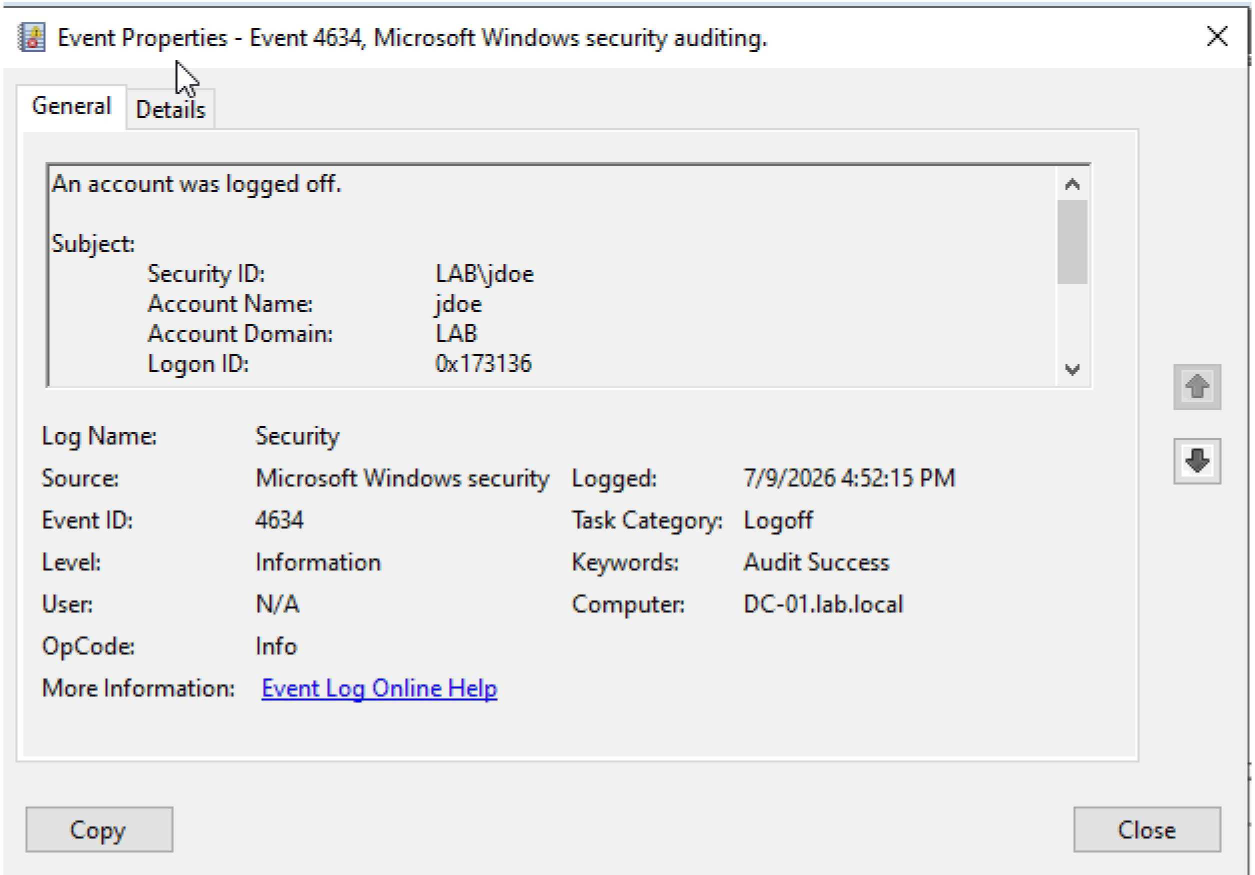


Group policy updated

17. Successfully enforced lockout policy by intentionally failing 5+ logon attempts:



Event Viewer security logs for Windows shows the audit logs for when the user failed to logon and for successful logon attempt:



Summary:

- Deployed Windows Server 2022 Domain Controller with Active Directory Domain Services in a virtualized lab environment
- Designed OU structure and implemented Role-Based Access Control (RBAC) with least-privilege user account management
- Authored and deployed Group Policy Objects enforcing password complexity, account lockout, and removable storage controls
- Configured Windows Security Auditing and analyzed Event Logs (IDs 4625, 4740) to document a simulated account lockout incident

Applications:

In this project, I simulated how an enterprise company grants access to their employees by using a VM to create a Domain Controller and Client to enforce Role-Based Access Controls based on the Organizational Units I created for each kind of user. I also configured Group Policy Objects to enforce password complexity, account lockout, and removable storage controls to increase the security posture for our domain, similar to how businesses reinforce their security across their entire deployment. I also went ahead and enabled Windows Security Auditing and analyzed Event Logs for logons, logoffs, failed attempts, and lockouts, just like how system administrators or SOC's may observe events for whenever there are security concerns.